MASSIMILIANO REMIGIO

Chemist, PhD · Materials and Surface Analysis · Analytical Methods and Data

+32 456 621 419 · Massimiliano.remigio97@gmail.com · Leuven, Belgium · Available immediately

PROFILE

Chemist completing a joint PhD at KU Leuven and Université libre de Bruxelles, with more than three years of daily hands-on lab work and earlier experience in industry. I characterise surfaces, thin films, and materials, develop and validate analytical methods, and model process data in Python. I am looking for R&D, analytical, process, or semiconductor materials roles.

CORE EXPERTISE

- XPS/UPS (UHV), STM/STS, AFM, C-AFM
- HPLC-MS · method development
- CVD and PVD · thin films
- DOE and Python · statistics, models
- SEM-EDX, XRPD, Raman, FT-IR, AFM-IR
- GC-MS, NMR, ICP-MS/OES (ppb to ppt)
- Cleanroom and etching · own substrates
- SOPs, deviation logs, CAPA records

EXPERIENCE

PhD Researcher, Analytical and Materials Chemistry

KU Leuven / Université libre de Bruxelles · *2022 to Present* · Leuven, Belgium

Thin films, surfaces, and molecular self-assembly, on a multi-instrument platform I run daily.

- **Surfaces and films:** Prepared films by CVD, PVD, and solution deposition and characterised them by XPS/UPS under UHV, STM/STS, AFM, SEM-EDX, and Raman mapping.
- **Analytical methods:** Developed HPLC-MS methods for purity determination and validated methods for linearity, precision, accuracy, and LOD/LOQ.
- **Data:** Planned multivariate DOE campaigns and built Python models linking process parameters and film structure to measured conductance.
- **Projects:** Ran three workstreams in parallel, coordinated campaigns with academic and industrial partners, and presented results to multidisciplinary audiences.

Publications: *Nature Materials (submitted)* · *J. Am. Chem. Soc. 2025* · *Cryst. Growth Des. 2025.*

MSc Researcher, Pilot-Scale Process Development

University of Bologna · *2019 to 2021* · Bologna, Italy

- **Pilot production:** Operated a pyrolysis pilot plant from commissioning through sustained production, defined its operating protocols, and troubleshot instabilities until the process was robust and repeatable.

Industrial QC and Materials Intern

Walter Tosto S.p.A. · *2017 to 2020* · Ortona, Italy

- **Quality and failures:** Qualified components against technical specifications in a certified production environment and investigated failures by SEM-EDX and XRPD.

Radiochemical Analyst, Nuclear Science Internship

Borexino Neutrino Experiment, INFN · *2016* · Gran Sasso, Italy

- **Trace analysis:** ICP-MS at ppb to ppt levels, with method validation and strict contamination control.

EDUCATION

PhD, Chemistry (Surface Science and Materials) KU Leuven and ULB · 2022 to Present
MSc, Chemistry (Organic and Green Chemistry) University of Bologna · 2019 to 2021
BSc, Chemistry (Materials and Chemistry) University of Bologna · 2016 to 2019

LANGUAGES

Italian Native · **English** Fluent · **Dutch** Actively developing, resident in Leuven · **French** Basic